AGENTIC RAG LEGAL CHALLENGE

STARTER GUIDE

09.03.2026

AGENDA

Today's plan

AGENTIC RAG LEGAL CHALLENGE

- What is ARLC
- Dataset structure
- Evaluation criteria
- Starter kit and Platform walkthrough
- Q&A Session

What is ARLC?

The **Agentic RAG Legal Challenge (ARLC)** is an international engineering competition for building production-grade legal AI systems.

The Goal

To benchmark Agentic RAG pipelines on realistic legal knowledge tasks under conditions close to real deployments.

The Process

Document Ingestion → Retrieval → Grounded Generation → Evaluation

KEY METRICS

- Answer Quality
- Latency (TTFT)
- Grounding
- Engineering Robustness

Official competition part of

Machines Can See Conference

Dubai AI Week

Principles

1. Real-World Relevance

The challenge is designed to resemble actual business AI systems, not toy ML tasks.

- Real document collections
- Multi-step pipelines
- Latency constraints
- Evaluation focused on grounded answers

2. Objective and Fair Benchmarking

At the same time, results must remain transparent, comparable, and reproducible.

- Structured dataset
- Automatic scoring
- Anti-gaming safeguards
- Leaderboard

MAIN PRIZES (BY TOTAL SCORE)

WINNER

1ST PLACE
$12,000

Grand Champion

2ND PLACE
$8,000

3RD PLACE
$4,000

SPECIAL NOMINATIONS

SPEED CHAMPION

$2,000

Fastest solution. Lowest avg TTFT (Total Score >= 70%)

EFFICIENCY EXPERT

$2,000

Most token-efficient. Highest Score/Token ratio (Total Score >= 70%)

RETRIEVAL MASTER

$2,000

Best grounding. Highest Grounding Score (Total Score >= 70%)

(2X) BEST PUBLICATION

$1,000

AI popularization. Best video/post/etc about the competition (by Jury's choice)

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BEST PUBLICATION CRITERIA

Methodological Rigor

The quality of the research methodology, including the correctness of experimental design and the validity of comparisons between approaches.

Insights on Building AI Systems

Practical lessons and insights derived from the work that inform how AI systems can be designed and implemented.

Error and Limitation Analysis

Clear understanding of where the system fails and why, including thoughtful analysis of limitations.

Reproducibility

Transparency of the approach, including the availability of code or sufficiently detailed methodological descriptions to enable replication.

Awards Ceremony

Online Conference in April

- Expert talks addressing the field of RAG and Legal AI
- Review of leading solutions, with expert commentary
- Awards ceremony of the challenge

Timing

Sign-up & Onboarding 11 February – 11 March

Public Release & Sample Questions 11 March – 18 March

Public dataset (80-100 docs), 100 sample questions, Platform live, leaderboard updates.

Full Corpus Unlocked 18 March

300 docs available for full evaluation (1,000 questions).

Submission Freeze 19 March 23:59

Validation & Review 20 March – 24 March

Final solutions validation and winner review period.

Dataset structure

```
n Name
- ----
2 01ab862cf9ef7012d76d06ebdafa1023ba91138961790824bbcdfd45d90adb86.pdf
2 02ec30ff6c98fa3adccd3834d73d76a2ae11dab91f2b4ea3501ff24db27436cc.pdf
9 03b621728fe29eb6113fcdb57f6458d793fd2d5c5b833ae26d40f04a29c85359.pdf
9 0471e83c1ea18086cfb6b3ff51da6f22b0efee337f10315b2593f782297ccb84.pdf
3 09660f78c26cd56c08c7253ed21ba01fb246092f482ccd8acd8e6f9b6fd2d917.pdf
1 0ee112f26f9b33c272a4fad301c52e7309c98236d627ad2451202ca6f5f0cdc7.pdf
5 1a255edc261961ec64870466a27ac4e25b5ebc2abe298e1b69f8dd2fc27288f6.pdf
1 1b446e196b4d1752241c8ff689a31ea705e98ad0c16b9d343c303664f72b64a1.pdf
1 2071fcaf840c53c30a7f4851ca1c0fc1a33475370170e8c22785f9e92d26608a.pdf
3 20be16a68c4d24768f569bcde5ddb29437204006c0c4262296c350b365672cd1.pdf
7 22442c5ee999e2519c68de908be511875a84f2b810ed540c2dcfcbcc65031434.pdf
4 2367e45430240e0210f1c54962e7e3a96b6b0908c56e01eb77ced99e114d61be.pdf
2 302a0bd8d67775e8dc5960ecec7879be566300d8b32c4b0153ba15ebdb279425.pdf
1 31af73c34191903fc3fab3233322ab3e97edb1820a966af313200cdb9bd3b24e.pdf
5 324b44750e9690ebf3204312bec975c1b1bca887c0c9d28666722d066e6ae114.pdf
3 33bc02044716acdfedb164b065bdaec098aaadcae863c591f9931c88e7307d16.pdf
1 3a574fc4f0baa5ec7ff0dcc822d7c55e4f9c493c24addc217c8df3d848c16fa5.pdf
```

```
[
  {
    "question": "Who were the claimants in case CFI (
    "answer_type": "names",
    "id": "cdddeb6a063f29cbea5f10b3dccbd83aa16849e1f3
  },
  {
    "question": "Summarize the court's final ruling i
    "answer_type": "free_text",
    "id": "6618184ee84fbebc360162dc3825868eec4e5e81aa
  },
  {
    "question": "Was the main claim or application in
    "answer_type": "boolean",
    "id": "df0f24b2b339c62162b82eb3add3a2a71a275ee768
  },
```

Dataset structure

Evaluation methodology and environment overview

Public Test Set

SIZE

100 questions on 87 documents

AVAILABILITY

Available now (Fixed March 11)

PURPOSE

Local pipeline debugging and validation of participant strategies

Private Test Set

SIZE

900 questions on the full corpus (~300 documents)

AVAILABILITY

Closed until the end of the competition

FINAL RANKING

Calculated after the submission deadline

PURPOSE

Protection against overfitting and ensuring fair ranking

~1000 Total Documents

1000 Total Questions

Answer Types - Deterministic (70%)

1 number

What is the state fee?

`1000`

What is the compensation amount?

`50000`

2 boolean

Does the document contain a Force Majeure definition?

`true`

Is unilateral client termination without cause allowed?

`false`

3 name

Who is appointed as arbitrator?

`"Chief Justice"`

Which court handles disputes?

`"DIFC Courts"`

4 names

What bodies can issue rules?

`["Chief Justice", "Board of Directors"]`

List applicable jurisdictions.

`["UAE", "UK"]`

5 date

When did the Employment Law 2019 come into force?

`"2019-02-08"`

Special answer: null

In all deterministic types, the answer `null` is allowed. This serves as recognition that the information is absent from the corpus and cannot be found or inferred.

Answer Types - Free text (30%)

free_text

FORMAT

Coherent text of 1–3 paragraphs, max 280 characters.

PURPOSE

Evaluate the model's ability to assimilate multiple facts into one coherent answer, structure an explanation, and match real assistant UX (not just listing facts, but also explanations).

Examples

"Formulate how the order describes the fee collection rules and the current rate for initiating proceedings."

Requirement: Focus on procedural finance details

"Explain under what conditions the client can terminate the contract and what consequences ensue."

Requirement: Focus on legal conditions and impact

Question types

QUESTION TYPE	COMMENT	PRIMARY ANSWER TYPE(S)
single-document	*"What is the notice period for termination?" → "30"; "Does the contract contain a force majeure clause?" → "yes"*	number, boolean, name, names
clause-analysis	*"What are Licensee's indemnification obligations under Section 8.2?" → (clause-analysis, free_text); "List exceptions to confidentiality" → (clause-analysis, names) → "auditors, regulators"*	free_text, names
multi-document	*"Compare liability limitations in Contract A and Contract B"*	free_text
negative	*"What is the penalty amount for NDA breach?"*	number, boolean, name, names, free_text
adversarial	*Based on the NDA, what is the personal liability of individual employees?*	free_text
uncertainty	*How is 'reasonable time' interpreted in this agreement?*	free_text

Dataset creation methodology

1. Corpus Collection
Compile a corpus of documents (laws, regulations, court decisions, etc.)

2. Question Generation
Generate Q&A pairs based on the documents

3. Automated Validation
An ensemble of LLMs evaluates each pair based on Context Relevance, Groundedness, and Answer Relevance.

- CONTEXT RELEVANCE
- GROUNDEDNESS
- ANSWER RELEVANCE

4. Expert Review
Professional lawyers manually verify the Q&A pairs.

5. Difficulty Calibration
Tested with a baseline RAG system to ensure a balanced set.

Evaluation

Deterministic metrics

Quantitative scoring for structured outputs and objective performance benchmarks.

Free text metrics

LLM-based evaluation for qualitative responses, checking tone and relevance.

Retrieval / Grounding metric

Applied across all question types to ensure accuracy and source adherence.

Time to First Token

Latency performance tracking for all question types to optimize user experience.

Evaluation — Deterministic metrics

TYPE	METHOD	FORMULA / LOGIC
number	Numerical tolerance	`abs(predicted - actual) < 0.01 * actual`
boolean	Exact match	—
name	Exact match	—
names	Jaccard index	`len(A ∩ B) / len(A ∪ B)`
date	Exact match (ISO 8601)	—
N/A (all types)	Exact match	`actual is None ? predicted is None`

EVALUATION — FREE TEXT METRICS

LLM Cascade: Gemini Flash 3, Claude Haiku → GPT 5.2

1 Correctness

Are all factual statements in the answer accurate and verifiable based on provided context?

Factual Verifiability

2 Completeness

Does the answer cover all key aspects and parts of the question?

Coverage Analysis

3 Grounding

Is every statement in the answer explicitly supported by information from context? Are there no hallucinations?

Contextual Adherence

4 Appropriate confidence

Is confidence adequately expressed? Does the system properly convey uncertainty if present?

Calibration

5 Clarity and relevance

Is the answer clear, concise, logically structured, and directly addressing the question posed?

Linguistic Quality

Σ AGGREGATE SCORE

$$\text{Judge_Score} = \frac{\Sigma(\text{count of 1s})}{\Sigma(\text{count of 1s} + \text{count of 0s})}$$

Evaluation — Grounding metrics

Measuring the factual alignment between retrieval results and the gold standard.

$$\text{Precision} = \frac{|\text{predicted} \cap \text{gold}|}{|\text{predicted}|}, \quad \text{Recall} = \frac{|\text{predicted} \cap \text{gold}|}{|\text{gold}|}$$

$$G = F_{\beta=2.5} = \frac{(1+\beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

Example: Gold set

`{(hashA, 1), (hashA, 2), (hashB, 5)} (3 pages)`

CASE	SUBMISSION PAGES	PRECISION	RECALL	SCORE
1. Perfect retrieval	`{(hashA,1), (hashA,2), (hashB,5)}`	1.0	1.0	1
2. Missing one page + noise	`{(hashA,1), (hashC,10)}`	0.5	0.33	≈ 0.35
3. Brute force (many extras)	`{(hashA,1), (hashA,2), (hashB,5), +4 irrelevant}`	≈ 0.43	1.0	≈ 0.88

The grounding metric evaluates how accurately the system references specific source pages from the gold set compared to the generated submission.

Evaluation — TTFT

TTFT	Modifier F
< 1.0 sec	**1.05**
1.0–2.0 sec	**1.02**
2.0–3.0 sec (Baseline)	**1.00**
3.0–5.0 sec	**0.95**
> 5.0 sec	**0.85**

Note: *TTFT is measured by participants and included in submissions. However, you have to submit the code for your solution, so your measurements will be checked.*

Final Evaluation

The Total Score is an aggregated metric computed from the averages of all child scores across the full submission. It is used solely for leaderboard ranking.

AGGREGATED SCORING FORMULA

$$\text{Total} = (0.7 \cdot S_{det_avg} + 0.3 \cdot S_{asst_avg}) \cdot G_{avg} \cdot T_{avg} \cdot F_{avg}$$

SYMBOL	COMPONENT	HOW COMPUTED	RANGE
`S_det_avg`	Deterministic accuracy	Average score across all deterministic questions	[0, 1]
`S_asst_avg`	LLM-as-Judge score	Average score across all free_text questions	[0, 1]
`G_avg`	Retrieval quality	Average G-Score (F-beta, β = 2.5) across all questions	[0, 1]
`T_avg`	Telemetry completeness	Average modifier across all questions in the submission	[0.90, 1.00]
`F_avg`	TTFT speed	Average modifier across all questions in the submission	[0.85, 1.05]

Platform & Starter Kit Demo

Q&A